

Exp. No. 1

Develop a lexical Analyzer to identify identifiers, constants, operators using C program.

main.c	Run ▶	Output
<pre>1 #include<stdio.h> 2 #include<ctype.h> 3 #include<string.h> 4 int main() 5 { 6 int i,ic=0,m,cc=0,oc=0,j; 7 char b[30],operators[30],identifiers[30],constants[30]; 8 printf("enter the string : "); 9 scanf("%[^\n]s",&b); 10 for(i=0;i<strlen(b);i++) 11 { 12 if(isspace(b[i])) 13 { 14 continue; 15 } 16 else if(isalpha(b[i])) 17 { 18 identifiers[ic] =b[i]; 19 ic++; 20 } 21 else if(isdigit(b[i])) 22 { 23 m=(b[i]-'0'); 24 i=i+1; 25 while(isdigit(b[i])) 26 {</pre>		<pre>enter the string : a = b + c * e + 100 identifiers : a b c e constants : 100 operators : = + * +</pre>
<pre>27 m=m*10 + (b[i]-'0'); 28 i++; 29 } 30 i=i-1; 31 constants[cc]=m; 32 cc++; 33 } 34 else 35 { 36 if(b[i]=='*') 37 { 38 operators[oc]='*'; 39 oc++; 40 } 41 else if(b[i]=='-') 42 { 43 operators[oc]='-'; 44 oc++; 45 } 46 else if(b[i]=='+') 47 { 48 operators[oc]='+'; 49 oc++; 50 } 51 else if(b[i]=='=') 52 {</pre>		<pre>55 } 56 } 57 } 58 printf(" identifiers : "); 59 for(j=0;j<ic;j++) 60 { 61 printf("%c ",identifiers[j]); 62 } 63 printf("\n constants : "); 64 for(j=0;j<cc;j++) 65 { 66 printf("%d ",constants[j]); 67 } 68 printf("\n operators : "); 69 for(j=0;j<oc;j++) 70 { 71 printf("%c ",operators[j]); 72 } 73 }</pre>